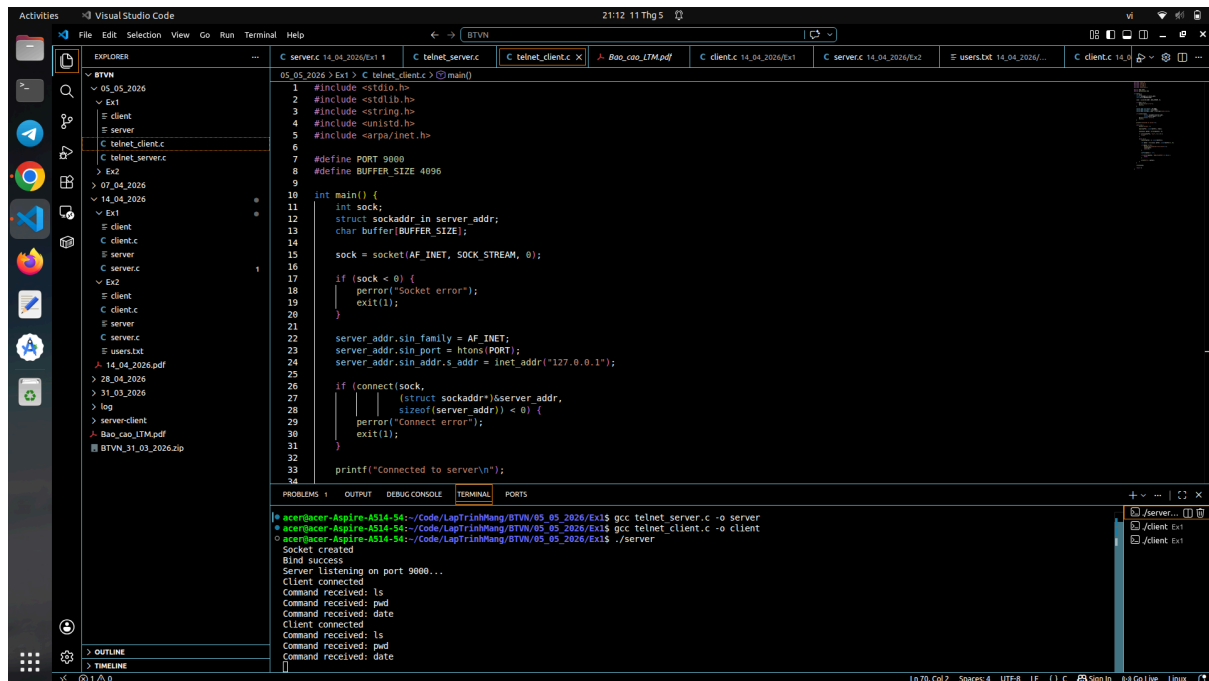


BÁO CÁO BÀI TẬP LẬP TRÌNH MẠNG

Đặng Thanh Phong - 210215627

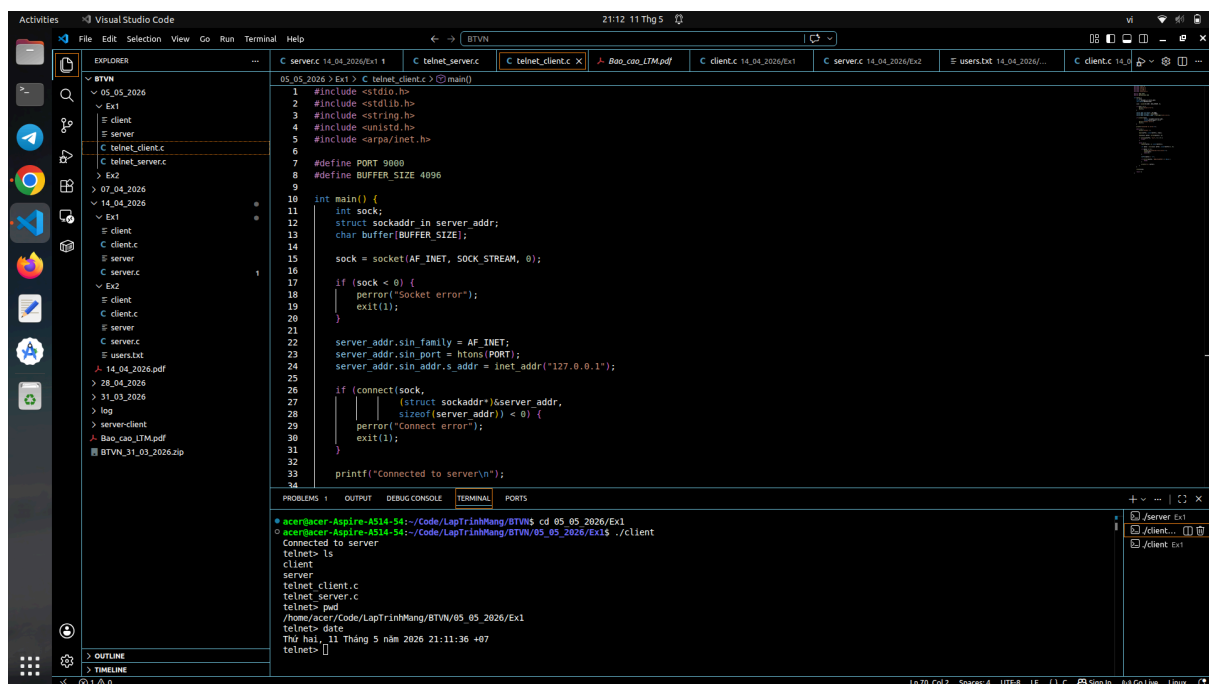
Ex1. Lập trình ứng dụng telnet_server sử dụng kỹ thuật multiprocessing



The screenshot shows the Visual Studio Code interface with the following components:

- Explorer:** Displays the file structure of the project, including folders for 05_05_2026, 14_04_2026, and 20_04_2026, and files like client.c, server.c, telnet_client.c, and users.txt.
- Code Editor:** Shows the source code for `telnet_client.c`. The code includes headers for `stdio.h`, `stdlib.h`, `string.h`, `unistd.h`, and `arpa/inet.h`. It defines `PORT 9000` and `BUFFER_SIZE 4096`. The `main` function sets up a socket, binds it to `127.0.0.1`, and listens for connections. When a connection is established, it prints "Connected to server\n".
- Terminal:** Shows the output of the compilation and execution. The commands executed are:
- `gcc telnet_server.c -o server`
- `gcc telnet_client.c -o client`
The output shows the server listening on port 9000 and receiving connections from the client.

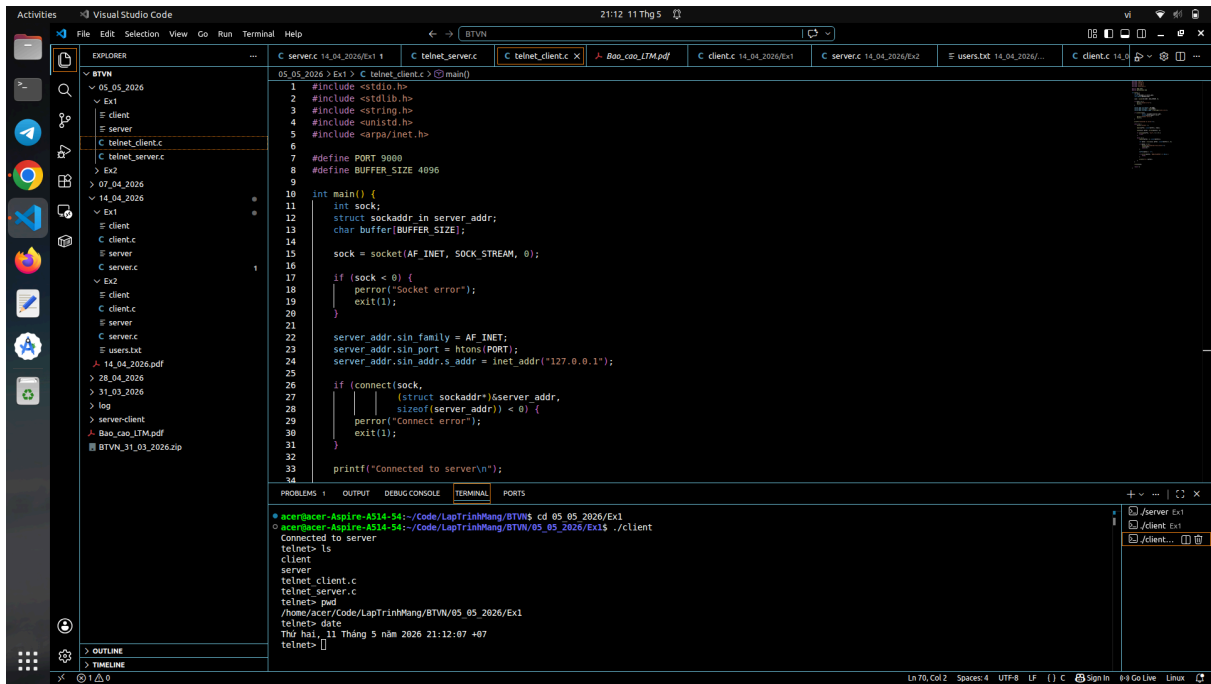
Hình 1. Hình ảnh chạy server



The screenshot shows the Visual Studio Code interface with the following components:

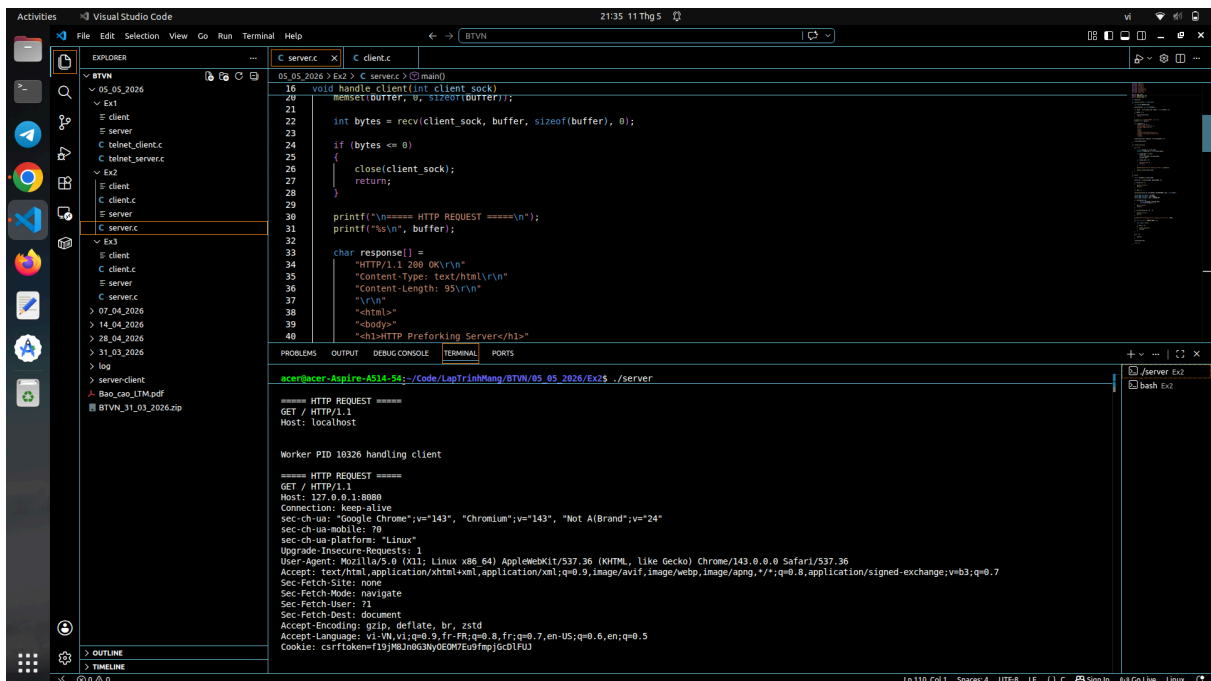
- Explorer:** Displays the file structure of the project, including folders for 05_05_2026, 14_04_2026, and 20_04_2026, and files like client.c, server.c, telnet_client.c, and users.txt.
- Code Editor:** Shows the source code for `telnet_client.c`. The code includes headers for `stdio.h`, `stdlib.h`, `string.h`, `unistd.h`, and `arpa/inet.h`. It defines `PORT 9000` and `BUFFER_SIZE 4096`. The `main` function sets up a socket, binds it to `127.0.0.1`, and listens for connections. When a connection is established, it prints "Connected to server\n".
- Terminal:** Shows the output of the compilation and execution. The commands executed are:
- `gcc telnet_server.c -o server`
- `gcc telnet_client.c -o client`
The output shows the server listening on port 9000 and receiving connections from the client.

Hình 2. Hình ảnh chạy client 1

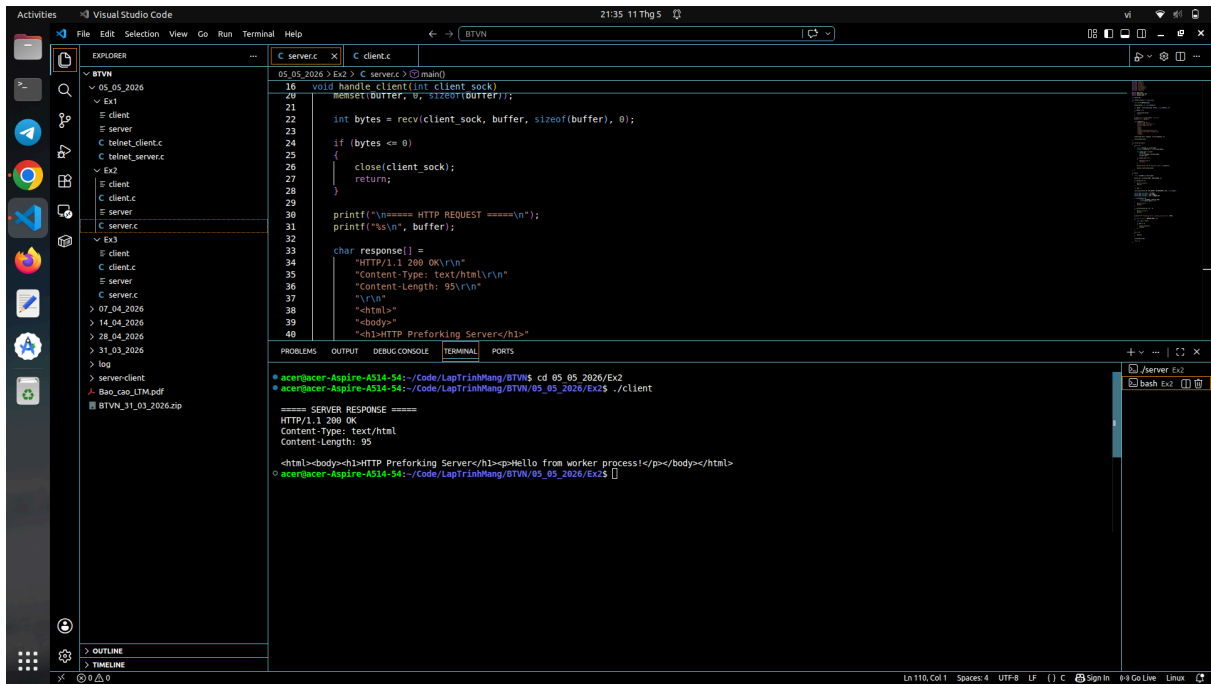


Hình 3. Hình ảnh chạy client 2

Ex2. Lập trình http_server dùng kỹ thuật preforking

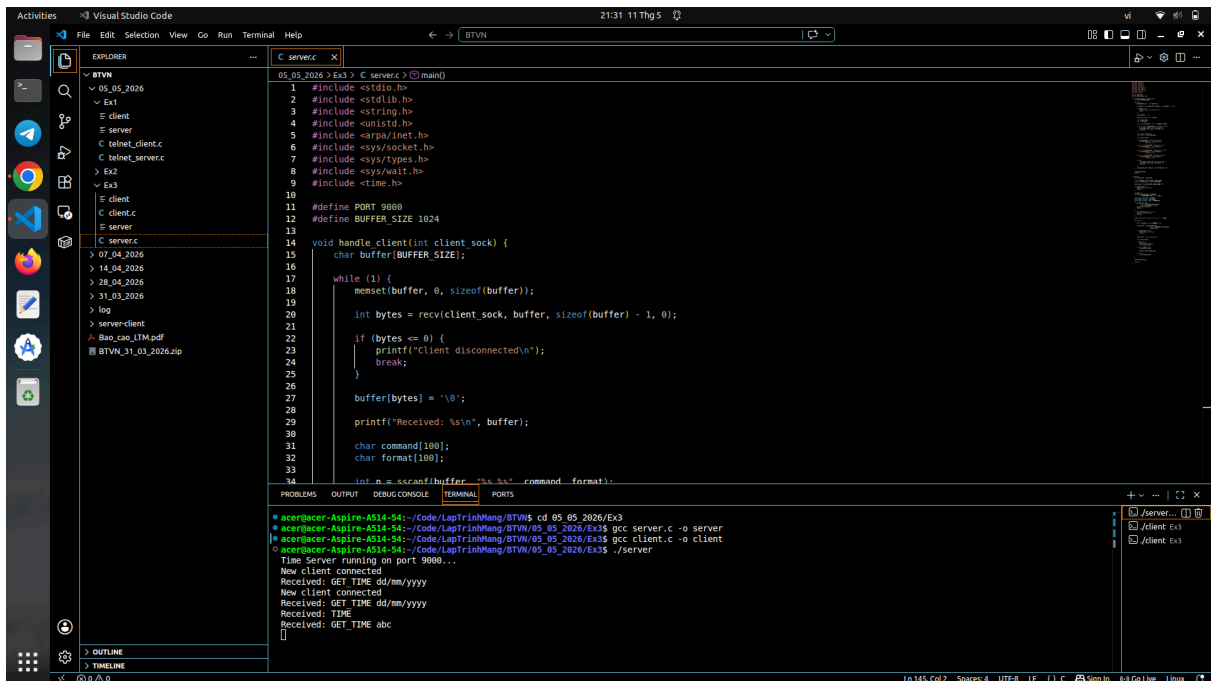


Hình 4. Hình ảnh chạy server

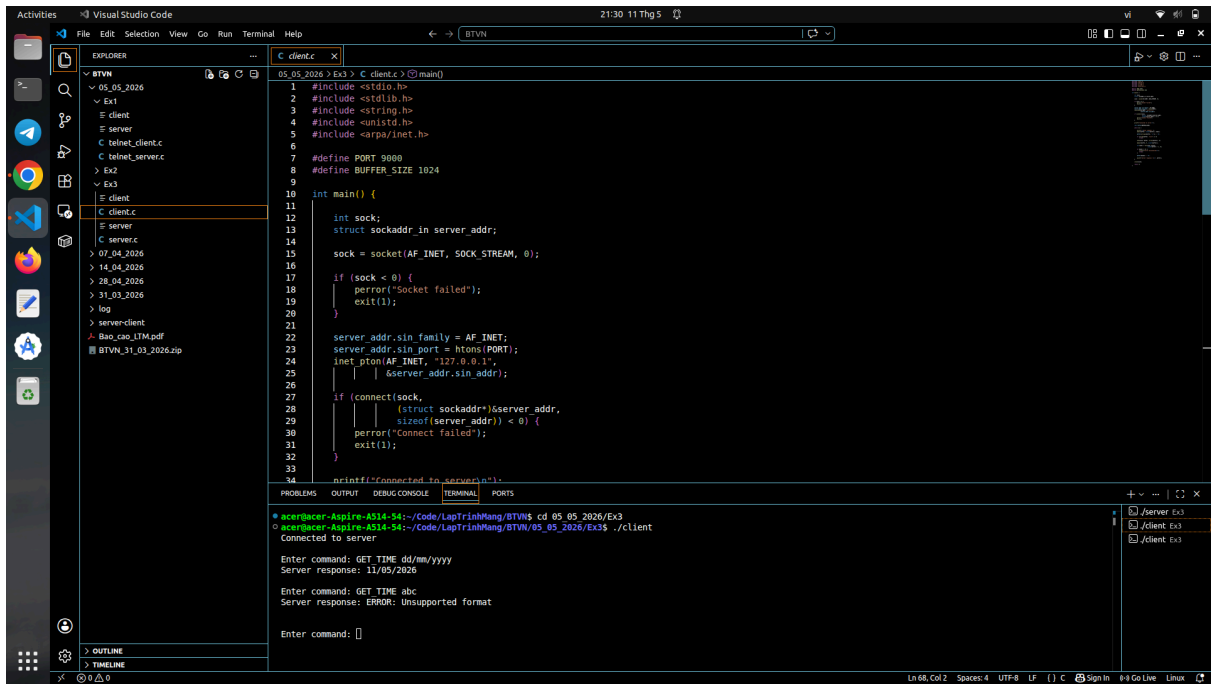


Hình 5. Hình ảnh chạy client

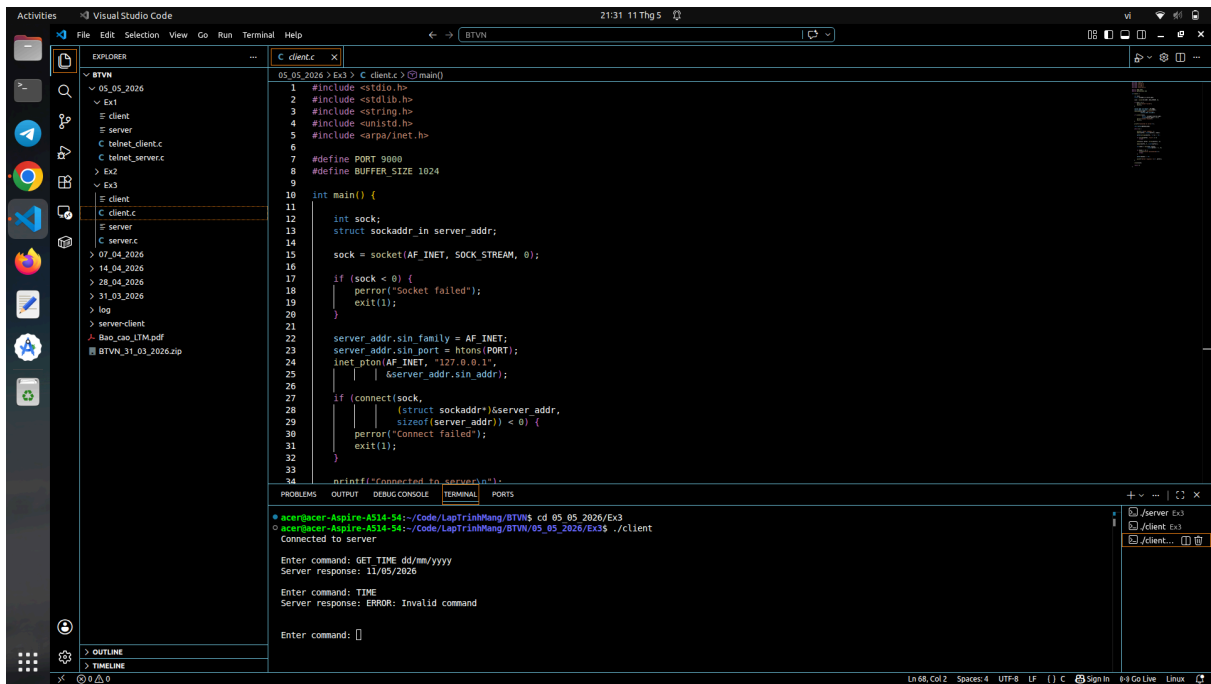
Ex3: Lập trình ứng dụng time_server



Hình 6. Hình ảnh chạy server



Hình 7. Hình ảnh chạy client 1



Hình 8. Hình ảnh chạy client 2